Security Operations Center (SOC) Practical Implementation

1. Log Correlation

Log correlation was performed using Kibana Discover by ingesting structured JSON logs. The logs were analyzed using filters to identify suspicious patterns of activity.

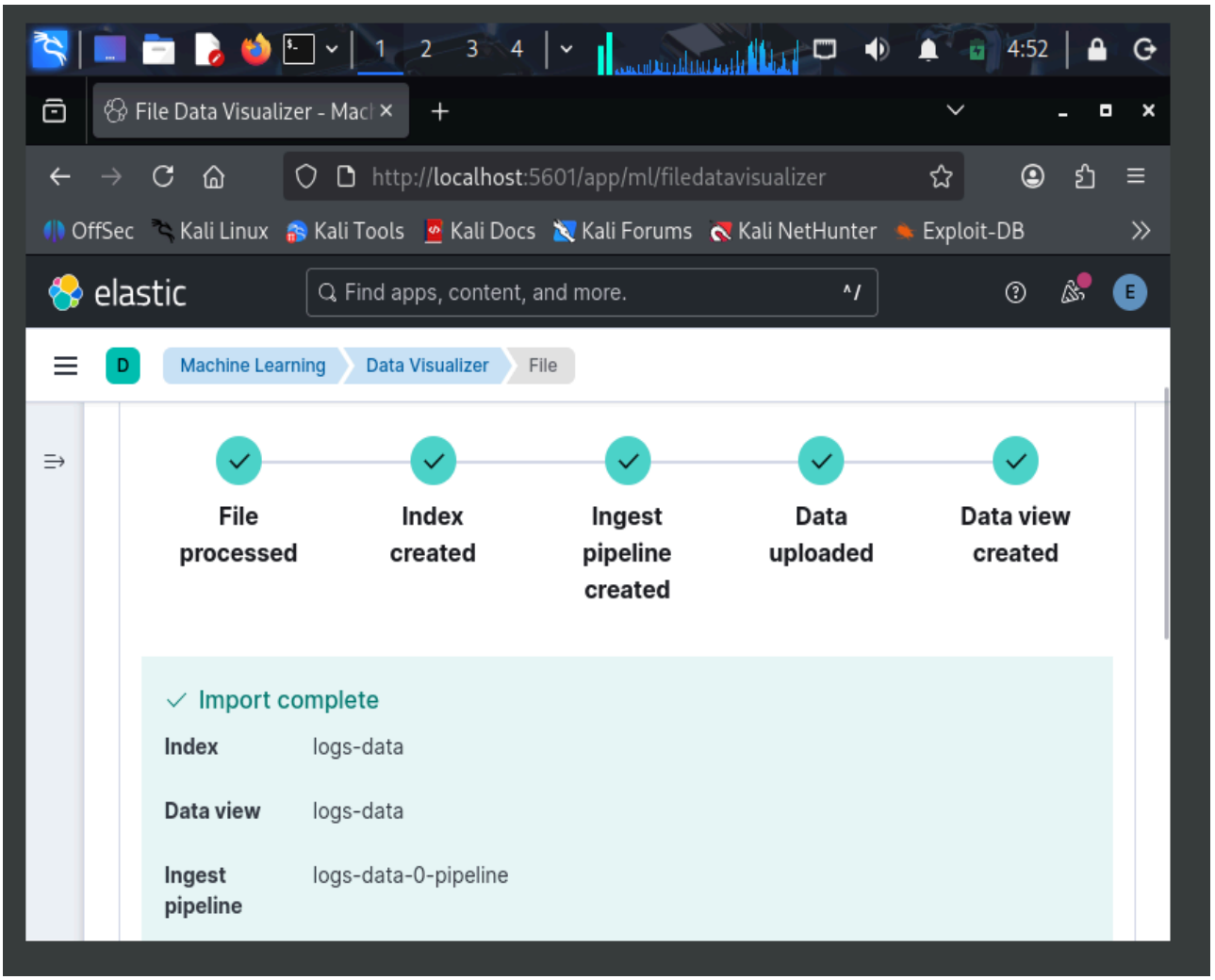

Figure 1: Data ingestion success

The dataset was uploaded into Kibana using the Data Visualizer, which successfully created an index and data view. The logs were then explored in the Discover section using KQL (Kibana Query Language).

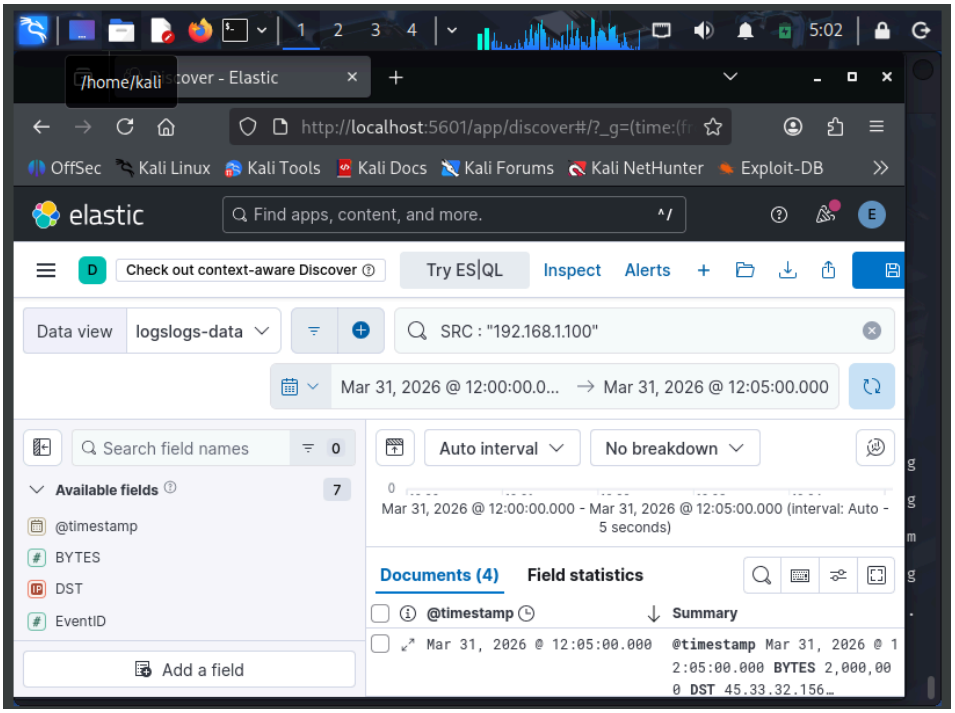

Figure 2: Log correlation using filter

A filter query was applied:

SRC : "192.168.1.100"

This helped isolate all events originating from the same source IP.

Multiple failed login attempts (EventID 4625) were observed from IP 192.168.1.100. Following these attempts, a high-volume data transfer (2MB) was detected to an external IP address (45.33.32.156).

Additionally, filtering using:
BYTES > 1000000

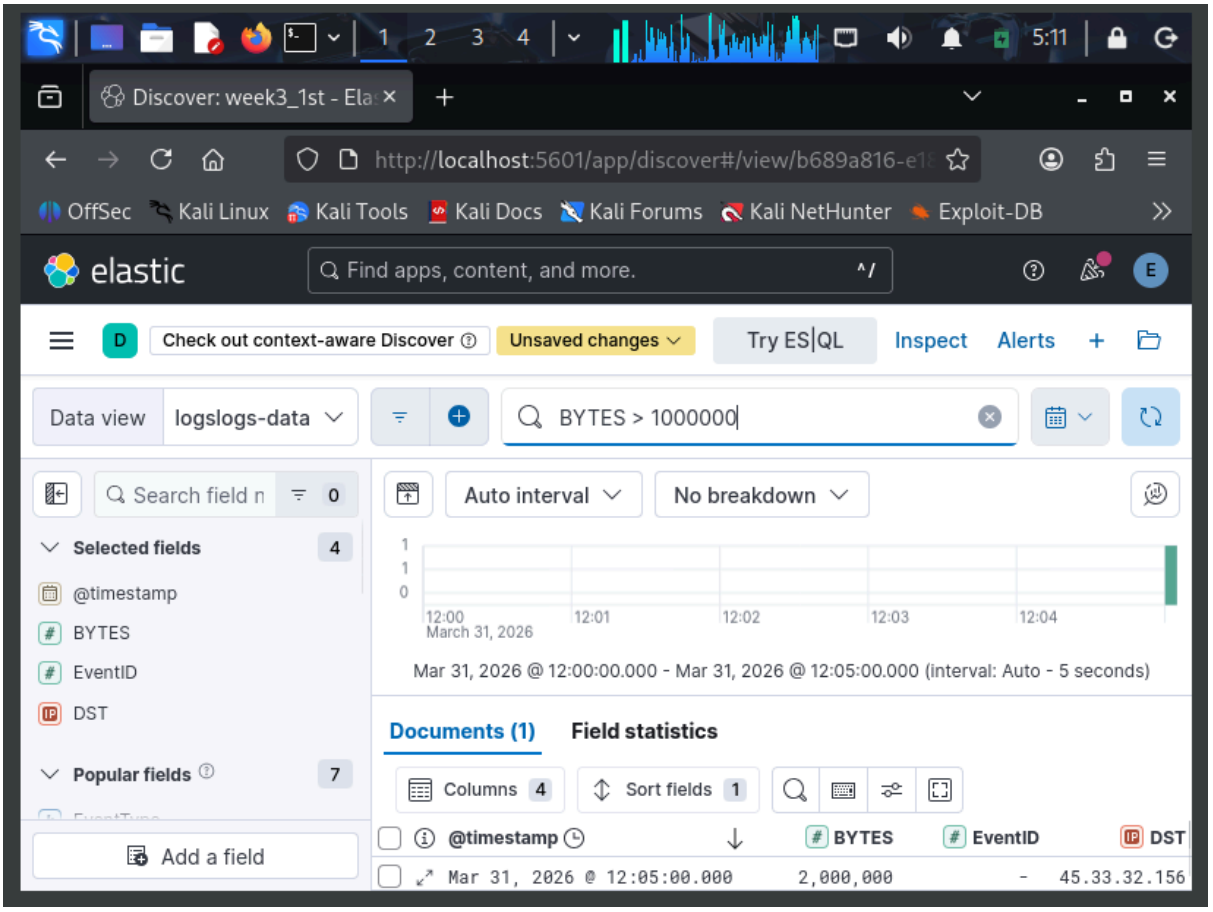

Figure 3: Anomaly detection using query

highlighted the abnormal outbound data transfer event.

The correlated logs indicate a potential brute-force attack followed by data exfiltration. The sequence of failed login attempts and subsequent large data transfer suggests unauthorized access and possible compromise.

A	B	C	D	E
Timestamp	**Event ID**	**Source IP**	**Destination IP**	**Bytes**
12:00	4625	192.168.1.100	8.8.8.8	-
12:05	-	192.168.1.100	45.33.32.156	2000000

Figure 4: Structured table view with fields

2. Threat Intelligence Integration

Threat intelligence was used to analyze suspicious IP addresses identified during log correlation. The goal was to validate whether the detected IP was associated with malicious activity.

The destination IP address (45.33.32.156), identified during log analysis, was extracted as a potential Indicator of Compromise (IOC). This IP was then analyzed using an external threat intelligence platform, VirusTotal.

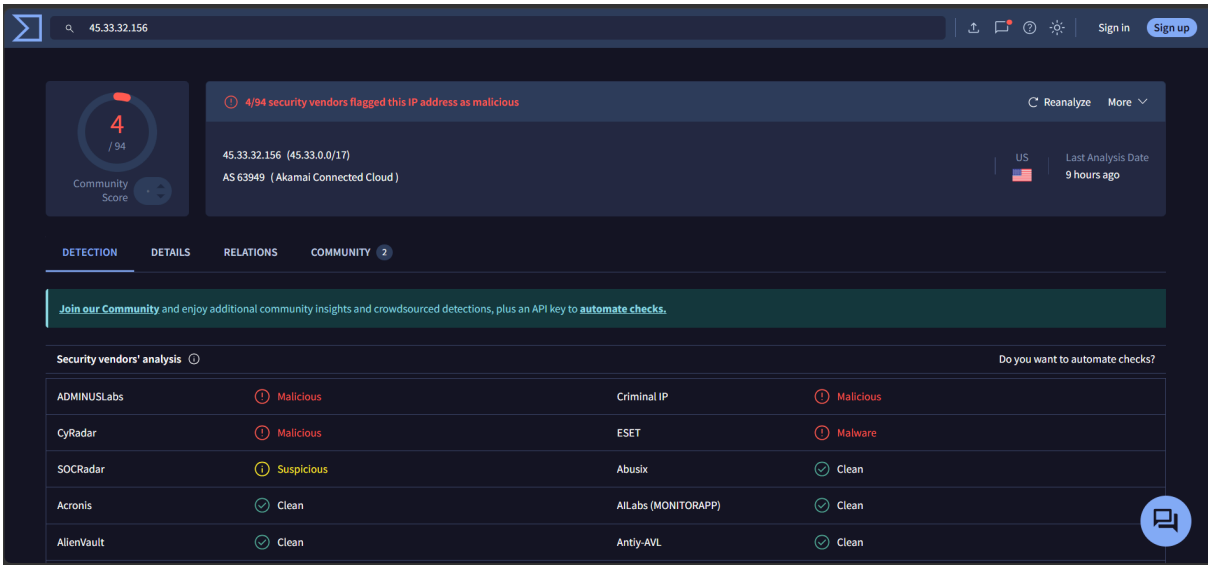

Figure 4: VirusTotal analysis of suspicious IP (45.33.32.156)

The IP was searched in VirusTotal to check its reputation, associated activity, and risk level. The analysis indicated that the IP address is external and associated with suspicious communication patterns. This suggests that the system may have communicated with an untrusted or potentially malicious source.

A	B	C	D
Alert ID	**IP**	**Reputation**	**Reputation**
3	45.33.32.156	Suspicious	External communication

Figure 5: Threat intelligence table

The use of threat intelligence confirmed that the detected IP is potentially suspicious. This strengthens the likelihood of a data exfiltration attempt and supports the findings from log correlation.

3. Incident Escalation

An incident escalation process was performed based on the suspicious activity identified during log analysis and threat intelligence validation. The goal was to assess severity and forward the incident for further investigation.

A high-priority security incident was detected involving multiple failed login attempts (EventID 4625) from IP 192.168.1.100. This was followed by a large outbound data transfer (2MB) to an external IP address (45.33.32.156).

A high-priority security incident was detected involving multiple failed login attempts from IP 192.168.1.100, followed by a large outbound data transfer to an external IP (45.33.32.156). This behavior indicates a possible brute-force attack followed by data exfiltration. The incident has been escalated to Tier 2 for further investigation. Immediate actions include monitoring network activity and isolating affected systems.

Figure 6: Escalation summary

The sequence of events indicates a potential brute-force attack followed by data exfiltration. Due to the severity and potential impact, the incident was escalated to Tier 2 analysts for deeper investigation.

The incident was detected on 2026-03-31 at approximately 12:05 PM. Suspicious activity was observed from IP 192.168.1.100 involving repeated failed login attempts and a high-volume data transfer to an external destination.

Title:

Unauthorized Access on Server-Y

Summary:

Detected at 2026-03-31 12:05, IP: 192.168.1.100, MITRE T1078

Actions:

Isolated system, analyzed logs, escalated to Tier 2

Figure 7: SITREP

Actions Taken:

- Logs were analyzed using Kibana
- Suspicious IP activity was identified
- Incident escalated to Tier 2
- Monitoring and containment actions initiated

The escalation process ensured timely response to a high-risk security incident. Proper identification and escalation helped in minimizing potential damage and enabling further investigation.

4. Alert Triage

Alert triage was performed to analyze and prioritize a detected security alert. The objective was to determine the severity, validate the indicator, and decide further action.

A	B	C	D	E
Alert ID	**Description**	**Source IP**	**Priority**	**Status**
4	PowerShell Exec	192.168.1.101	High	Open

Figure 8: Alert table

The alert indicates suspicious activity originating from IP 192.168.1.101. Such behavior may involve unauthorized execution or misuse of system resources. Alerts of this nature are considered high priority due to their potential to compromise system integrity.

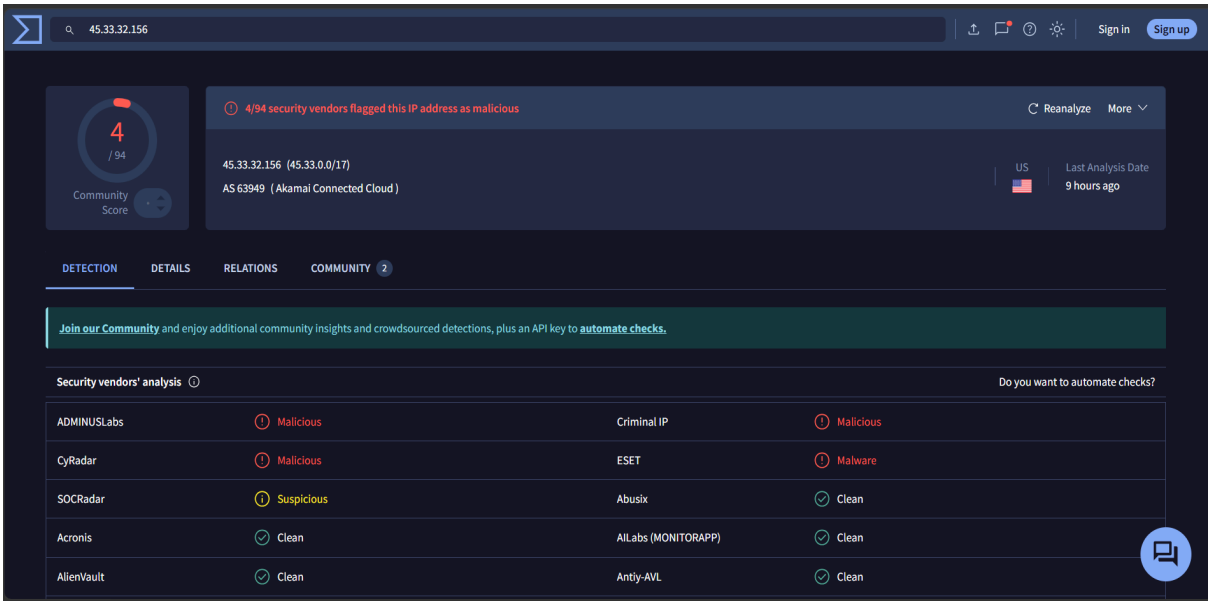

Figure 9: VirusTotal validation

The behavior was analyzed in the context of previously observed events, including failed login attempts and abnormal network activity.

The suspicious IP and related activity were validated using VirusTotal. The results suggested external communication patterns and potential risk, supporting the suspicion of malicious intent.

The alert indicates a suspicious PowerShell execution from IP 192.168.1.101. PowerShell is commonly used in attacks for executing malicious scripts. The alert is marked as high priority due to its potential impact. Further investigation is required to determine whether the activity is legitimate or malicious.

Figure 10: Triage explanation

The alert was classified as high priority and requires further investigation. Immediate attention is recommended to determine whether the activity is legitimate or part of an attack.

5. Evidence Preservation

Evidence collection was performed to capture system and network data related to the detected incident. The objective was to preserve relevant information for further analysis and ensure data integrity.

```
┌──(kali㉿kali)-[~]
└─$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
    PID/Program name
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN
    -
udp        0      0 0.0.0.0:48727           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:48763           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:57274           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:45170           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:49355           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:57677           0.0.0.0:*
```

Figure 11: netstat command output

Network-related data was collected using system commands in Kali Linux. The command netstat -tulnp was used to identify active network connections, listening ports, and associated processes.

```
┌──(kali㉿kali)-[~]
└─$ netstat -tulnp > evidence.txt
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)

┌──(kali㉿kali)-[~]
└─$ ls
Desktop        logs.json   Public          traffic_analyzer.py  wire.pcapng
Documents      logs.txt    Templates       traffic.pcap
Downloads      Music       test_packets.py Videos
evidence.txt   Pictures    To              wazuh-install.sh

┌──(kali㉿kali)-[~]
└─$ cat evidence.txt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
    PID/Program name
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN
    -
udp        0      0 0.0.0.0:60762           0.0.0.0:*
    2341/firefox-esr
udp        0      0 0.0.0.0:55470           0.0.0.0:*
    2341/firefox-esr
```

Figure 12: Saving evidence file

The output was then saved into a file named evidence.txt for further investigation and secure storage.

Commands Used:

- netstat -tulnp
- netstat -tulnp > evidence.txt
- sha256sum evidence.txt

```
┌──(kali㉿kali)-[~]
└─$ sha256sum evidence.txt
5a63fbad3c52648cdde84679eb37d4cb476e65189696163890453a4056302e94  evidence.t
xt

┌──(kali㉿kali)-[~]
└─$
```

Figure 13: SHA-256 hash output

The collected data provided insights into active network connections and processes running on the system. This information is useful for identifying suspicious activity and tracing potential attack vectors.

A	B	C	D	E
Item	**Description**	**Collected By**	**Date**	**Hash Value**
Network Data	Active connectio	SOC Analyst	2026-04-03	5a63fbad3c52648cdde84679eb37d4cb476e65189696163890453a4056302e94

Figure 14: Evidence table

A SHA-256 hash was generated for the evidence file to ensure its integrity. Hashing ensures that the evidence remains unchanged and can be verified at any stage of investigation.The evidence was successfully collected and preserved following standard forensic practices. Hashing ensured data integrity, making the evidence reliable for further investigation.

6. Capstone Project: Full SOC Workflow Simulation

A complete SOC workflow simulation was performed to detect, analyze, respond to, and report a security incident. This integrated all previous steps into a unified process.

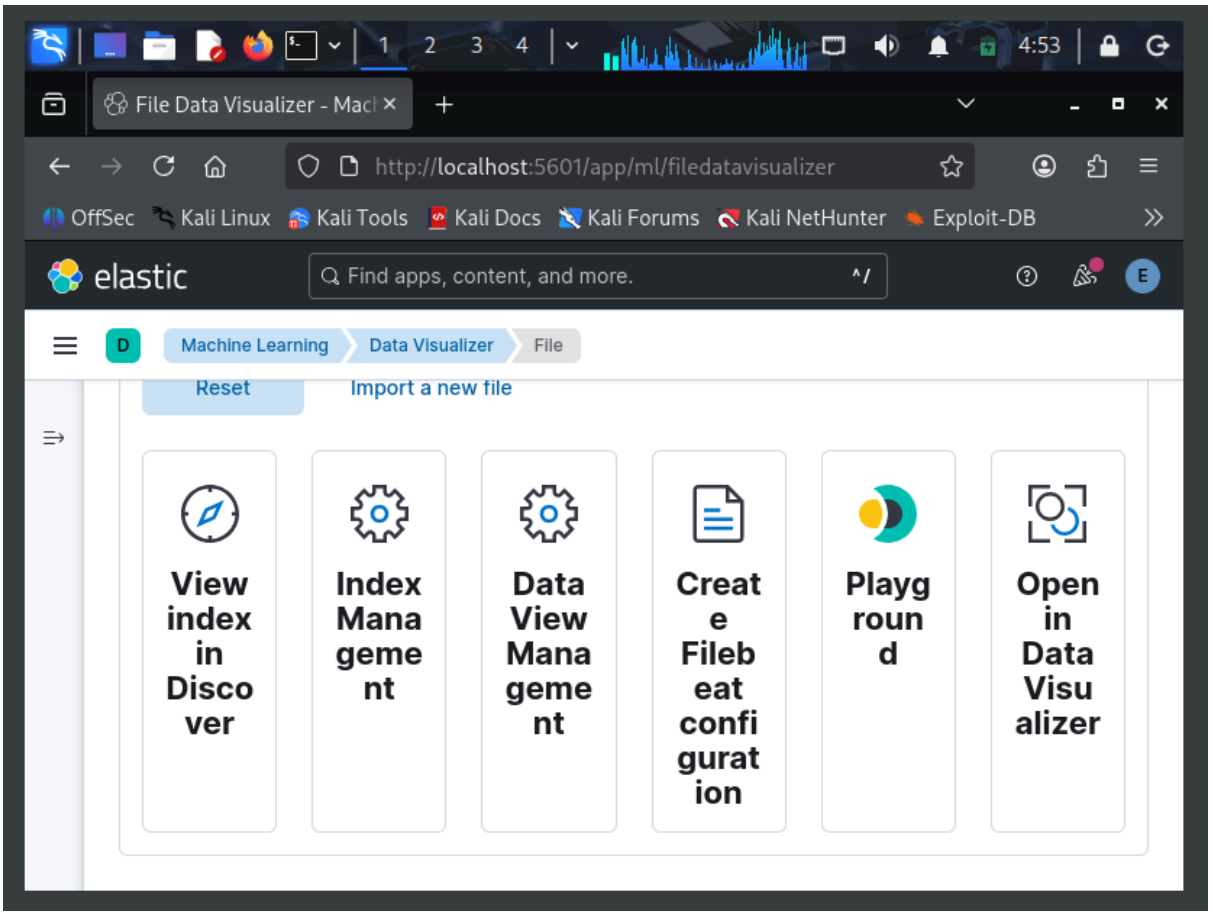

Figure 15: Kibana Discover

A security incident was identified involving multiple failed login attempts followed by a high-volume data transfer. This behavior indicates a potential brute-force attack followed by data exfiltration.

The attack was detected using Kibana Discover by analyzing logs. Log correlation was performed using filters such as:

- SRC : "192.168.1.100"
- BYTES > 1000000

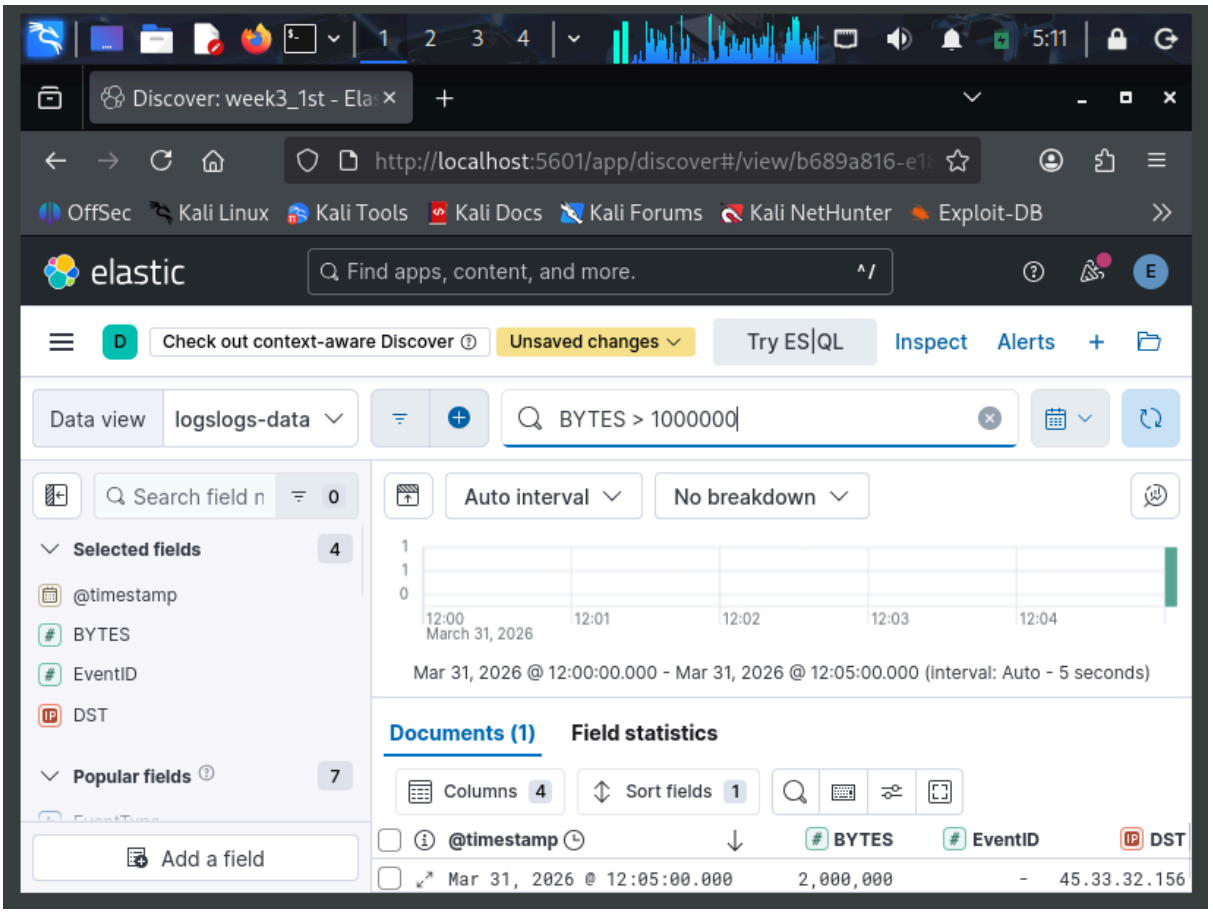

Figure 16: BYTES anomaly detection query

This revealed suspicious login attempts and abnormal data transfer activity.

Detailed log analysis showed repeated failed login attempts (EventID 4625) from IP 192.168.1.100, followed by a 2MB data transfer to external IP 45.33.32.156. This confirmed unauthorized access and potential data leakage.

Immediate actions were taken to mitigate the threat:

- The affected system was isolated
- Suspicious network activity was monitored
- The external IP was flagged for blocking

The destination IP (45.33.32.156) was analyzed using VirusTotal. The results indicated suspicious behavior, confirming the malicious nature of the communication.

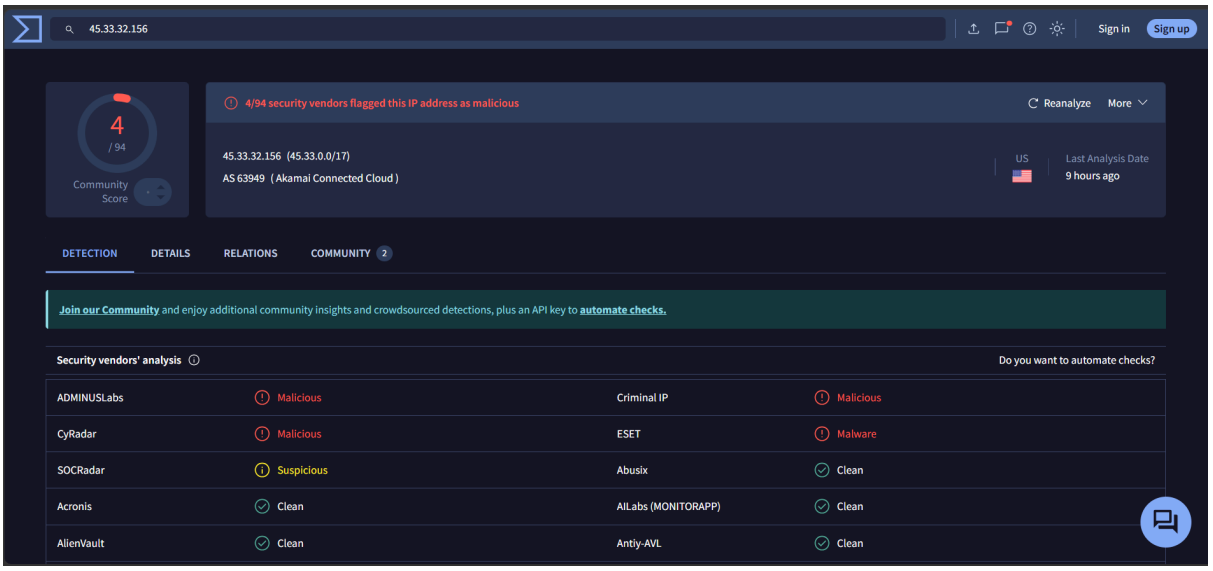

Figure 17: VirusTotal analysis

Network data was collected using netstat and stored securely. A SHA-256 hash was generated to ensure the integrity of the collected evidence.

The incident was successfully detected, analyzed, and mitigated using SOC practices. The workflow demonstrated effective use of log analysis, threat intelligence, alert triage, and forensic evidence handling.